



# Macro Forecasting with the TFT and Fed Speeches

Minna Heim, Chris Truill and Anna Zeitz



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- Macro forecasts matter for
  - Governments
  - Financial markets
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- The Federal Reserve (Fed) has a **dual mandate**:
  1. Price stability
  2. Maximum employment
- Accurate forecasts of inflation, unemployment, and GDP are crucial for interest rate decisions

**Can we do better than standard macro models?**

# Research Question

- The Fed communicates with the public through official statements and speeches

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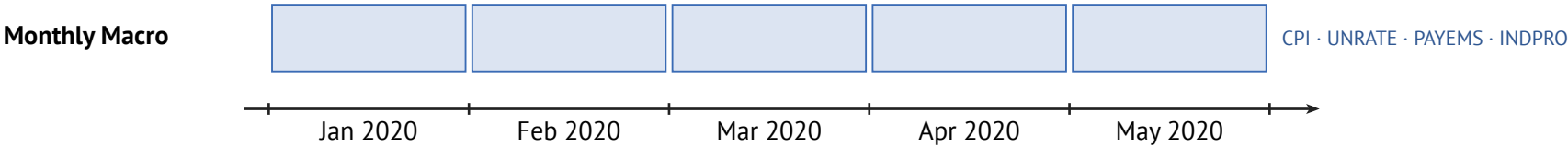
- The Fed communicates with the public through official statements and speeches
- It may communicate additional information:
  1. **Superior information**: private data and internal models not available to the public
  2. **Forward guidance**: signals about future policy to guide market expectations



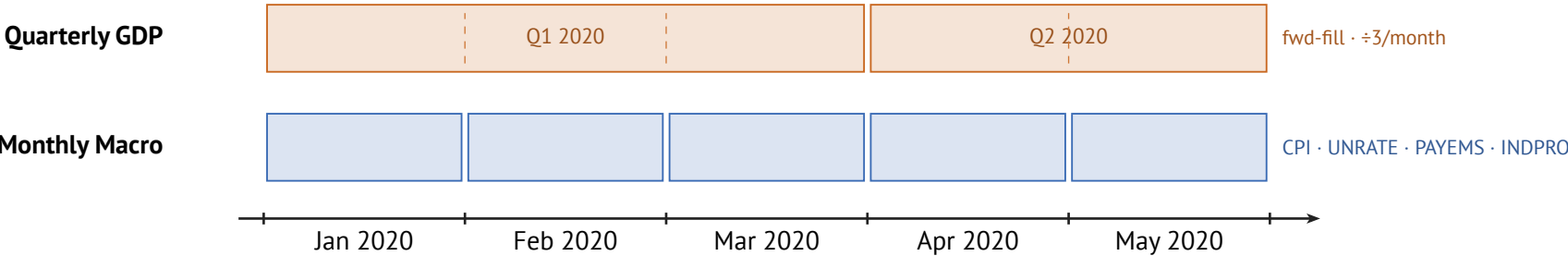
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  - It may communicate additional information:
    1. **Superior information**: private data and internal models not available to the public
    2. **Forward guidance**: signals about future policy to guide market expectations
- ⇒ **Do Fed speeches contain information useful to forecast macroeconomic indicators?**

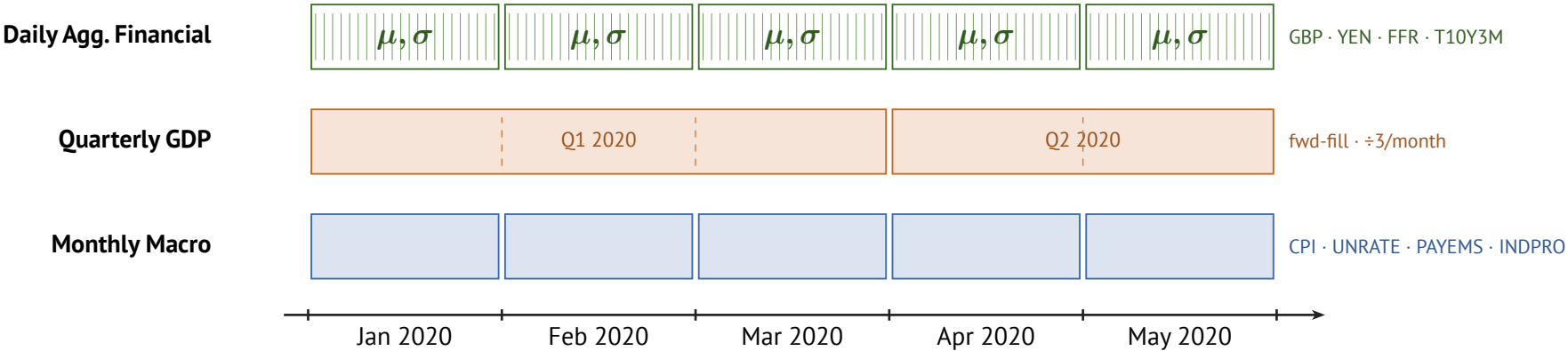
# Data Alignment



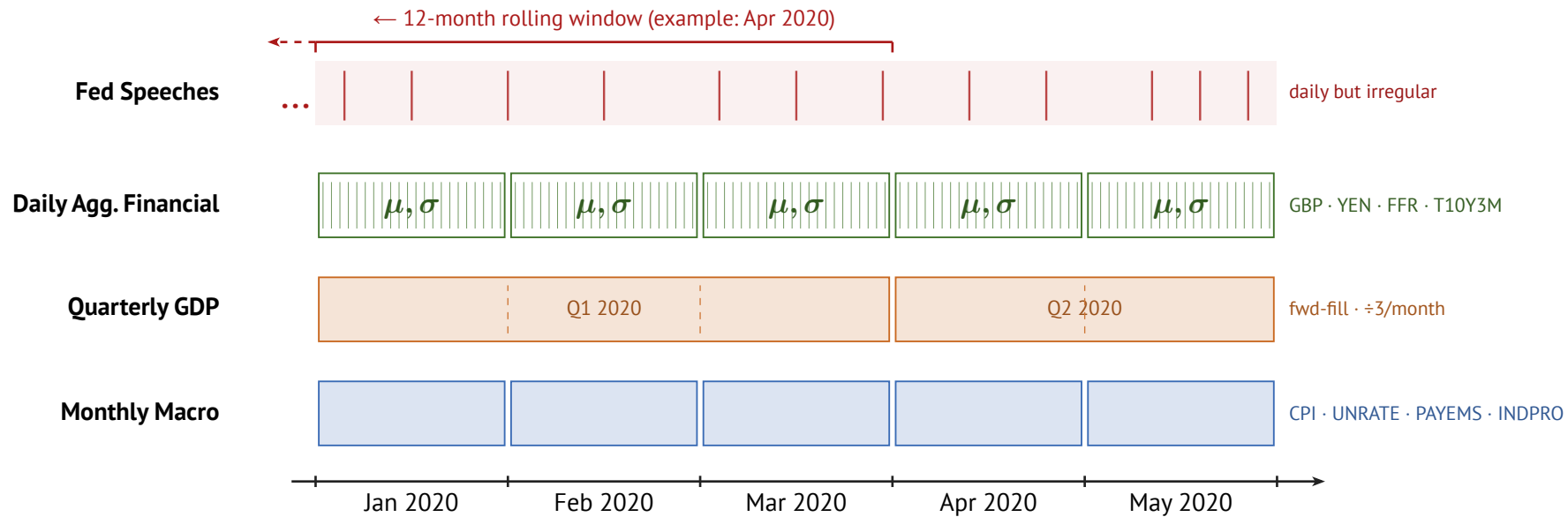
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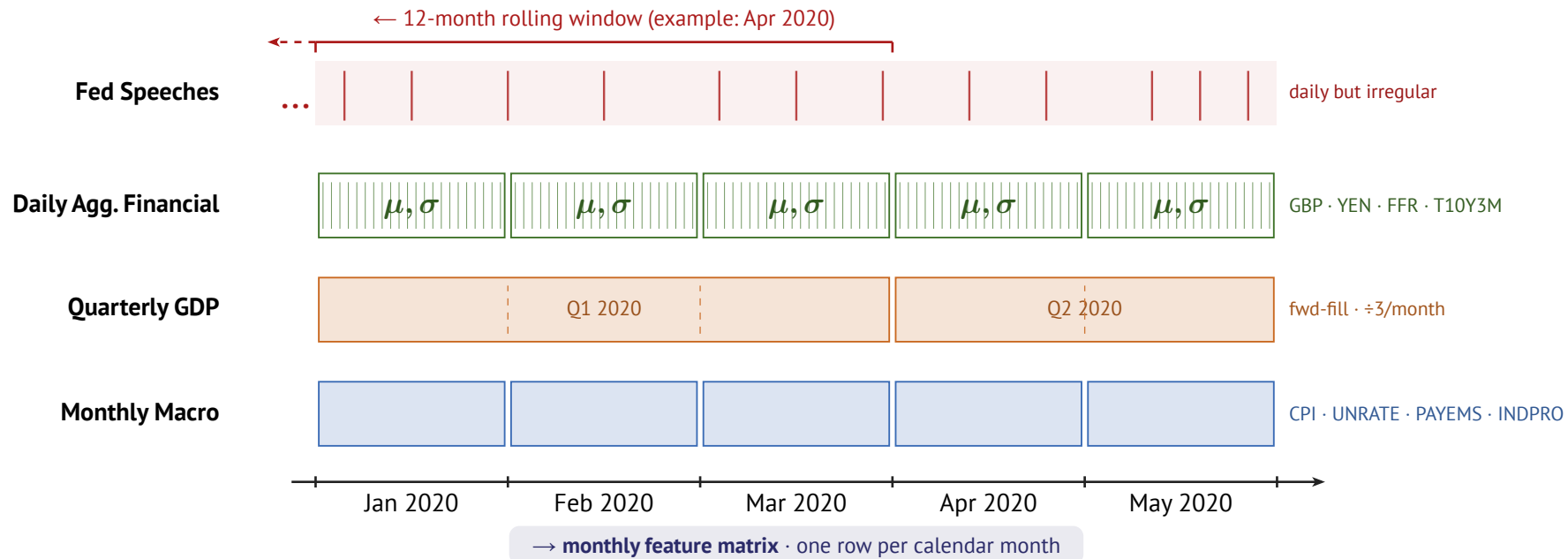
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# Models

# Benchmarks

AR(p)

$$y_t = c + \sum_{i=1}^p \varphi_i y_{t-i} + \varepsilon_t$$

ARIMA(p, d, q)

$$\Delta^d y_t = c + \sum_{i=1}^p \varphi_i \Delta^d y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t$$

■ Integration:  $d$ -th difference of  $y_t$       ■ MA component:  $q$  lags of residuals  $\varepsilon_t$



# Temporal Fusion Transformer

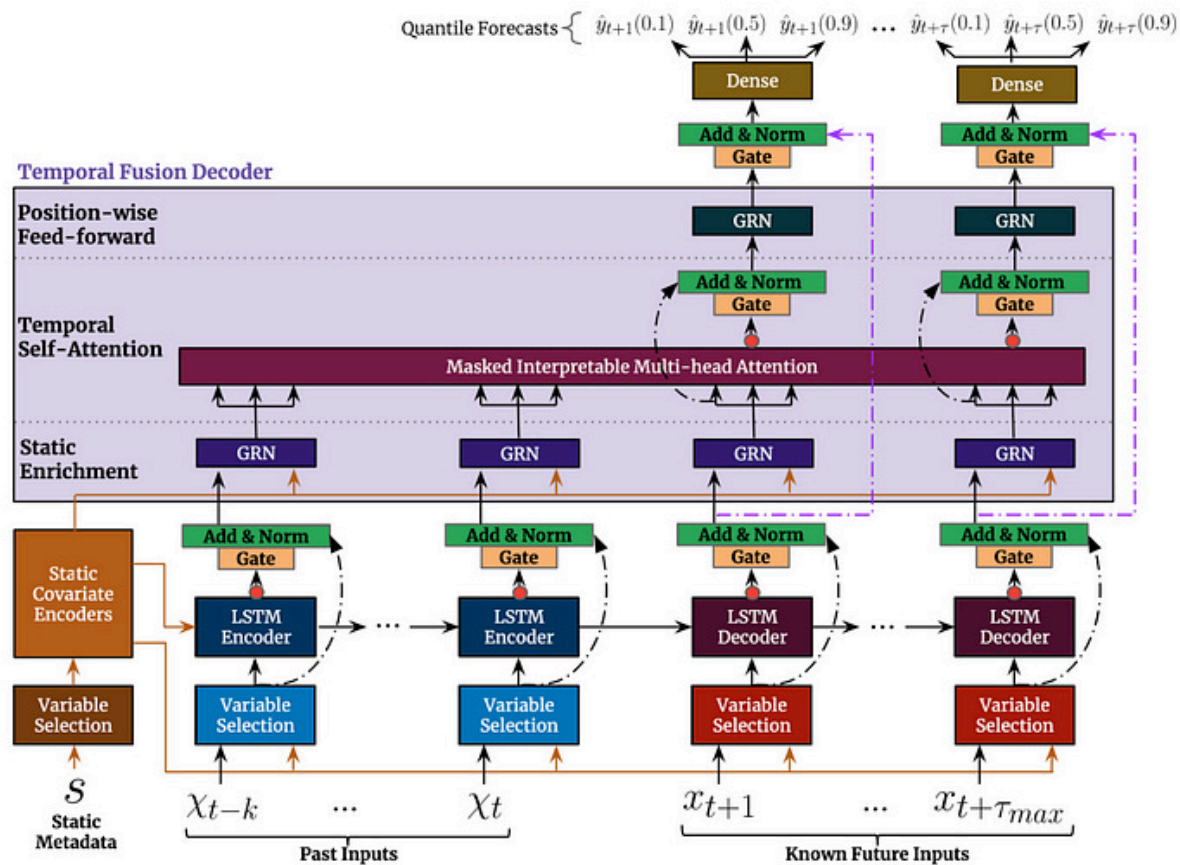


Figure 1: Model Architecture of the TFT (Lim et al., 2021)

# Speech Embeddings

# Speech Embeddings

Two models chosen for complementary perspectives:

1. **FinBERT**: BERT-based, pre-trained on financial statements (Yang et al., 2020)
2. **FOMC-RoBERTa**: fine-tuned on FOMC data, hawkish/dovish classification (Shah et al., 2023)

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**Processing**: chunk-and-average (512-token windows, 128-token overlap)

- Assumption: information spread across entire speech, not just the beginning

**Dimensionality reduction**:  $768 \Rightarrow X$  dims via Principal Component Analysis or Factor Analysis

- $X$  is treated as hyperparameter

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Speeches are held at **irregular** and **daily** frequency: how to align with monthly horizon?

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  - More recent speeches receive higher weight
  - $\lambda$  treated as fixed; speech window  $X$  tuned as hyperparameter

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3. **Attention-based**: key = embedding (after dimensionality reduction), query = mean macro state
  - Given current macro conditions, which past speeches are most informative?
  - Fitted on training data only  $\Rightarrow$  no leakage!



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All methods also track: voter speech ratio, chair speech ratio, gender share

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Tune in two stages:

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## 1. **Stage 1: Architecture (Macro-Only TFT)**

- Bayesian search via Optuna (TPE sampler)
- 2-fold CV, up to 50 trials
- Includes: encoder length, hidden size, normalizer

# Hyperparameter Tuning

Tune in two stages:

## 1. **Stage 1: Architecture (Macro-Only TFT)**

- Bayesian search via Optuna (TPE sampler)
- 2-fold CV, up to 50 trials
- Includes: encoder length, hidden size, normalizer

## 2. **Stage 2: Speech Embedding Params**

- Architecture fixed from Stage 1
- Tune: aggregation, reduction, PCA dims, speech window

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  - Context: training data + model's own previous predictions
  - Same assumption for AR, ARIMA and TFT  $\Rightarrow$  fair comparison

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$\Rightarrow$  All models evaluated on the same information set



# Results

## Holdout Evaluation

# Holdout Predictions ( $h=12$ )

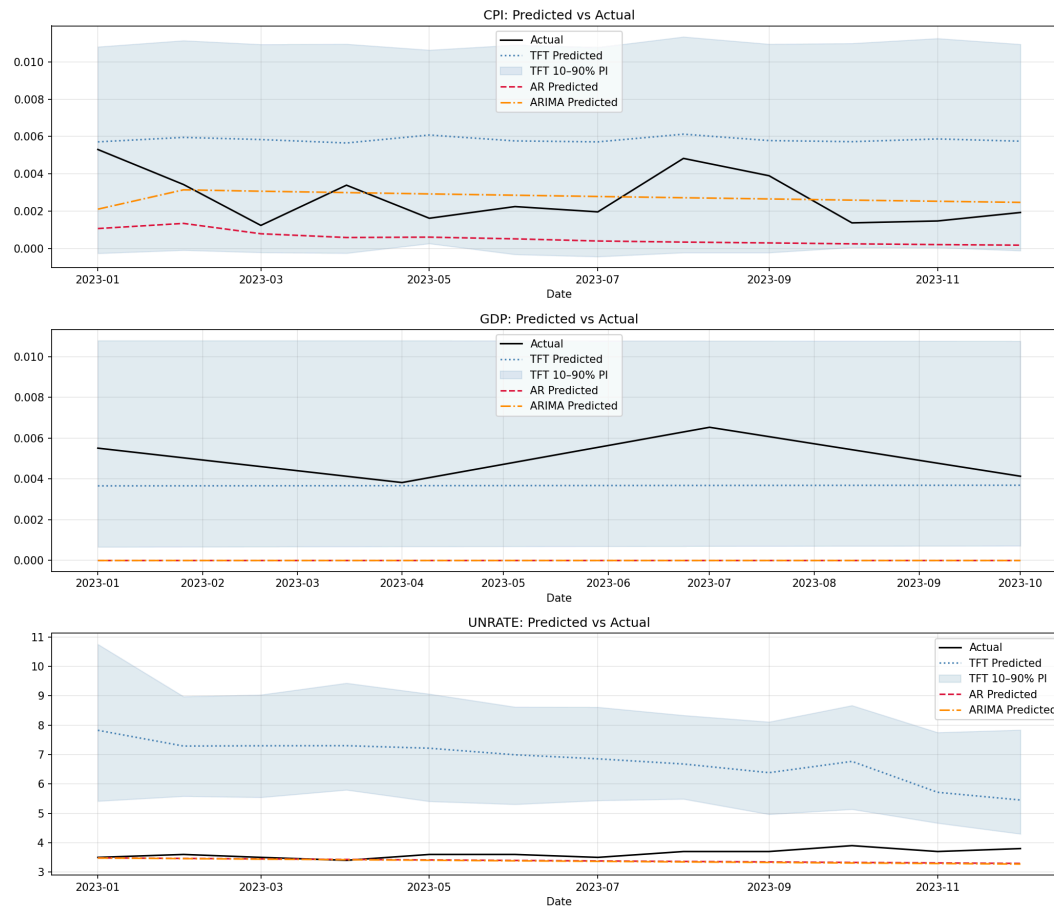


Figure 2: Holdout Predictions  $h=12$ , Macro Only

# Holdout Predictions ( $h=12$ )

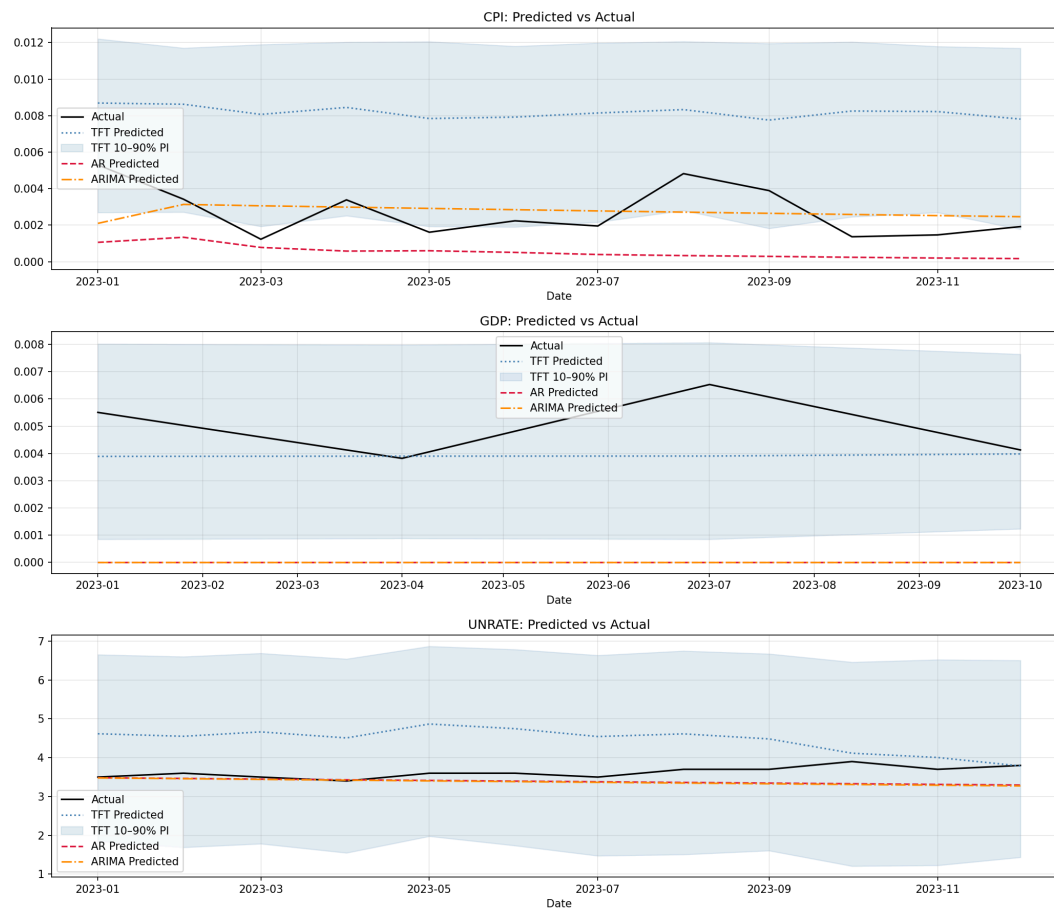


Figure 3: Holdout Predictions  $h=12$ , Including Embeddings

# Holdout Predictions ( $h=6$ )

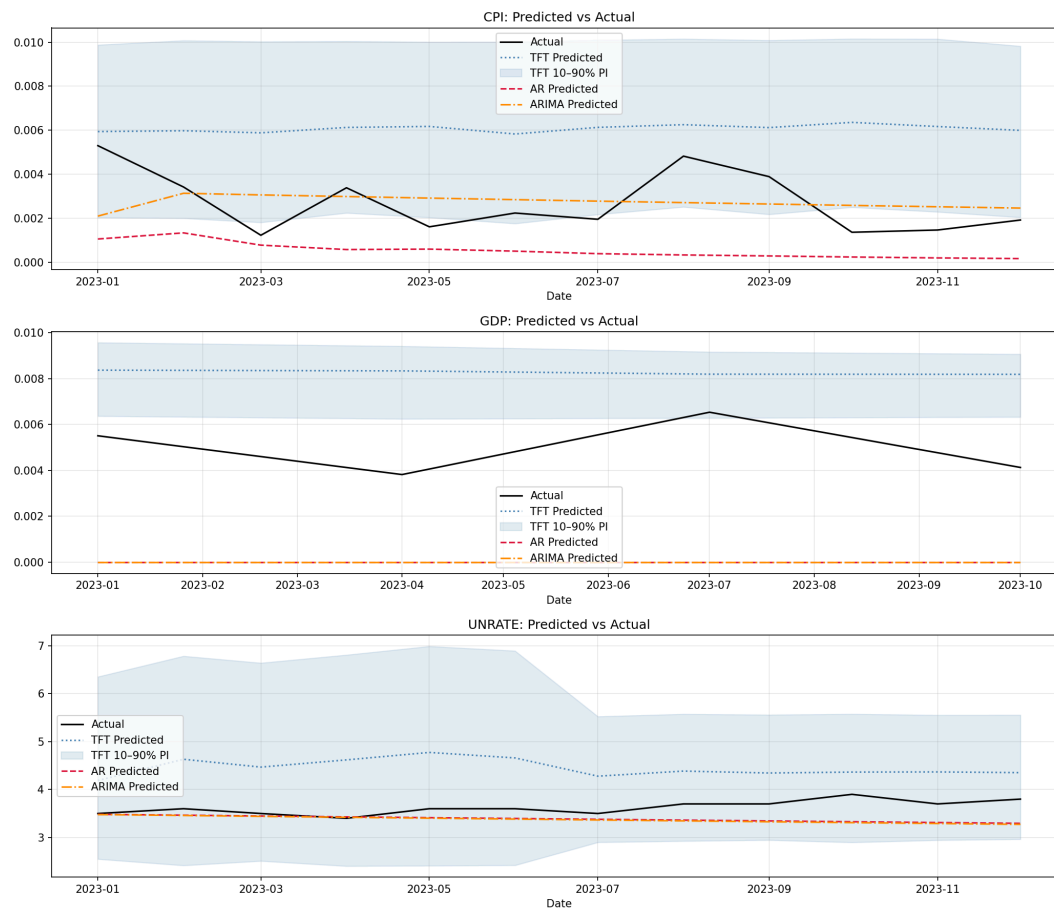


Figure 4: Holdout Predictions  $h=6$ , Macro Only

# Holdout Predictions ( $h=6$ )

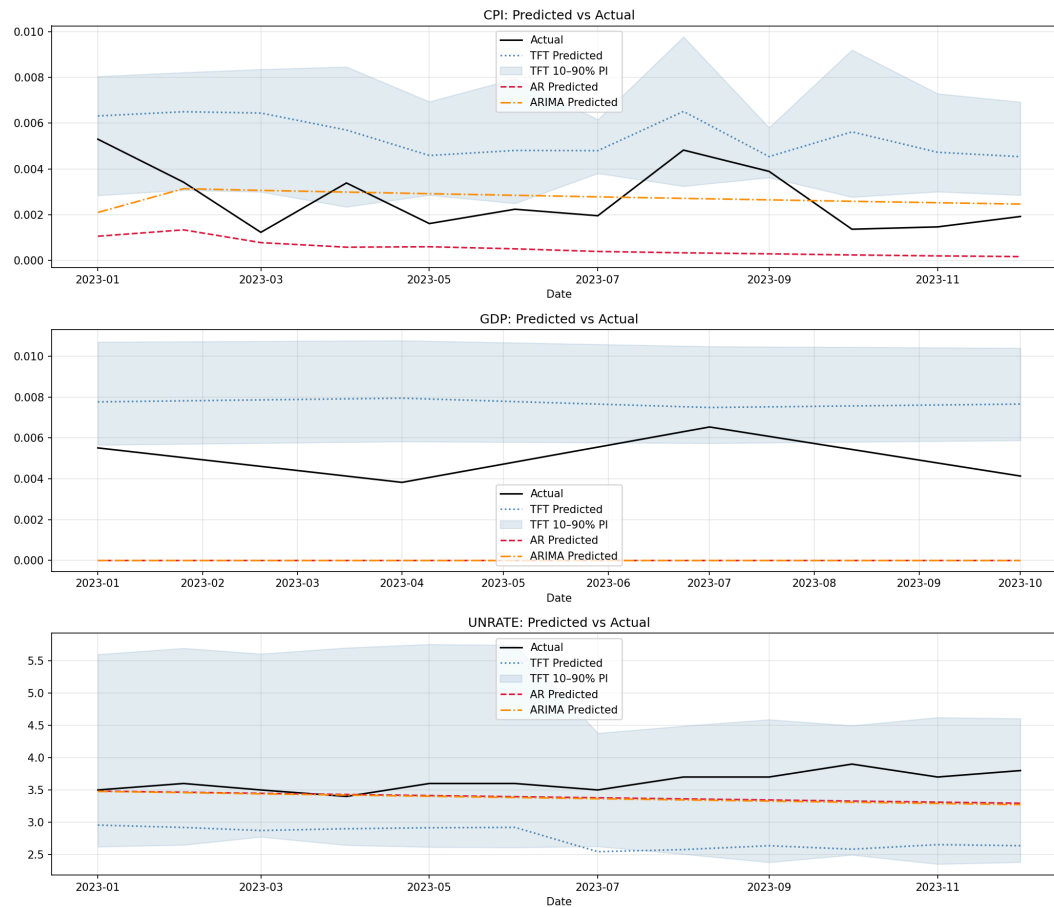


Figure 5: Holdout Predictions  $h=6$ , Including Embeddings

# Holdout Predictions ( $h=3$ )

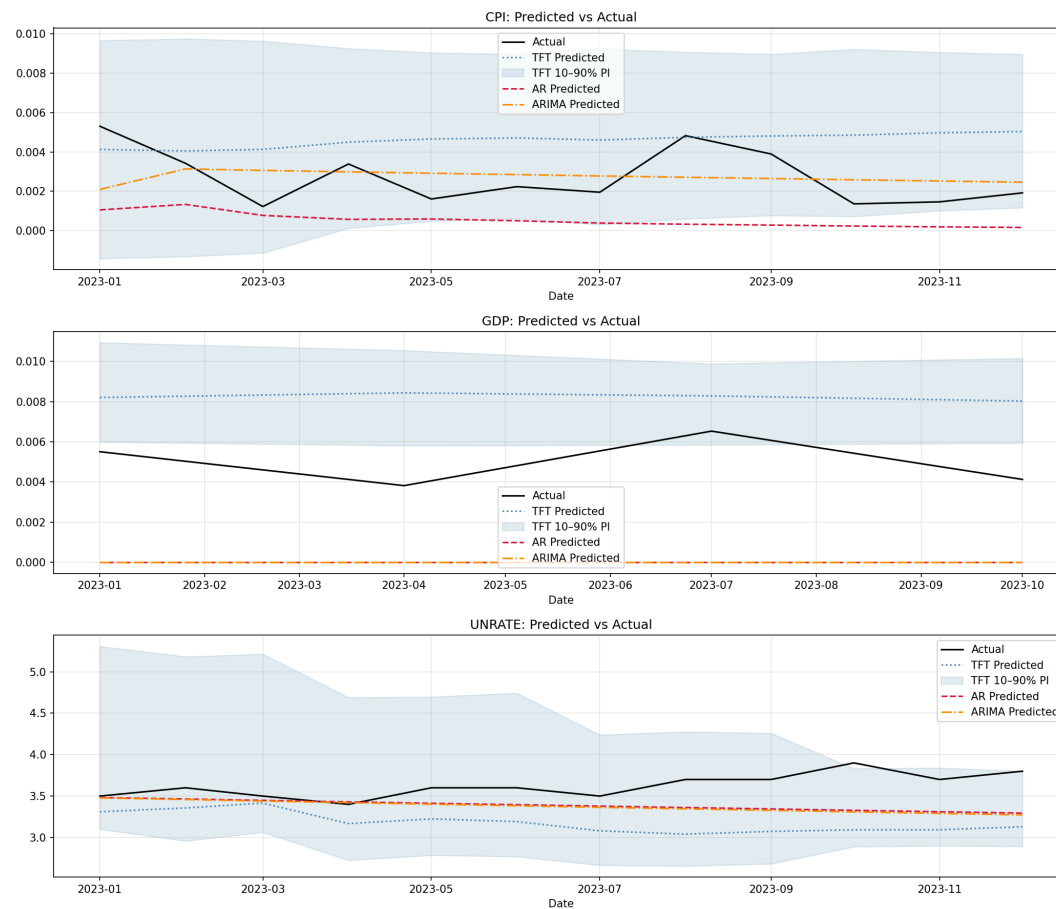


Figure 6: Holdout Predictions  $h=3$ , Macro Only

# Holdout Predictions ( $h=3$ )

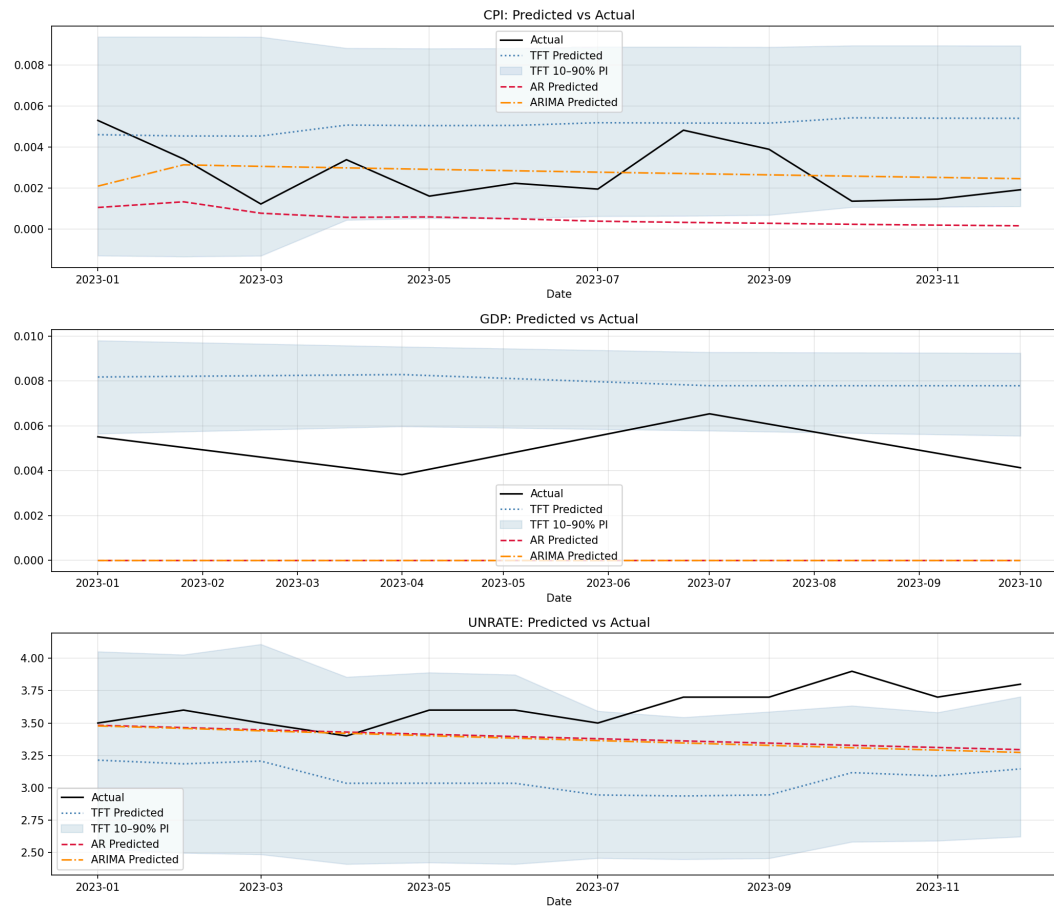


Figure 7: Holdout Predictions  $h=3$ , Including Embeddings

# Holdout Metrics: All Models & Horizons

| $h$       | Model   | CPI            |                | GDP            |                | UNRATE         |               |
|-----------|---------|----------------|----------------|----------------|----------------|----------------|---------------|
|           |         | MAE ↓          | RMSE ↓         | MAE ↓          | RMSE ↓         | MAE ↓          | RMSE ↓        |
| <b>3</b>  | AR      | 0.00218        | 0.00252        | 0.005          | 0.00511        | <b>0.24238</b> | <b>0.3012</b> |
|           | ARIMA   | <b>0.00122</b> | <b>0.00146</b> | 0.005          | 0.00511        | 0.25415        | 0.31466       |
|           | TFT     | 0.00209        | 0.00239        | 0.00324        | 0.00343        | 0.44346        | 0.49451       |
|           | TFT+Emb | 0.00245        | 0.00276        | <b>0.00301</b> | <b>0.00324</b> | 0.55122        | 0.57663       |
| <b>6</b>  | AR      | 0.00218        | 0.00252        | 0.005          | 0.00511        | <b>0.24238</b> | <b>0.3012</b> |
|           | ARIMA   | <b>0.00122</b> | <b>0.00146</b> | 0.005          | 0.00511        | 0.25415        | 0.31466       |
|           | TFT     | 0.00336        | 0.00363        | 0.00327        | 0.00345        | 0.82816        | 0.86254       |
|           | TFT+Emb | 0.00271        | 0.00296        | <b>0.00271</b> | <b>0.00298</b> | 0.86496        | 0.9042        |
| <b>12</b> | AR      | 0.00218        | 0.00252        | 0.005          | 0.00511        | <b>0.24238</b> | <b>0.3012</b> |
|           | ARIMA   | <b>0.00122</b> | <b>0.00146</b> | 0.005          | 0.00511        | 0.25415        | 0.31466       |
|           | TFT     | 0.00311        | 0.00339        | 0.00133        | 0.00172        | 3.18856        | 3.27634       |
|           | TFT+Emb | 0.00546        | 0.00559        | <b>0.00112</b> | <b>0.00155</b> | 0.83562        | 0.92786       |



# Results

- Incl. Tables, Figures

# Robustness Checks

- Shuffling Embeddings
- Using unrelated text (German Texts of Franz Kafka)
- Use not full speech but only first 512 tokens of speech???

# Future Work

- Hyperparameter tuning: switch to one stage tuning! Also: context-dependent attention
- Explore New Methods:
  - use Foundational Model (i.e. TabPFN, TimesFN)
  - Try out SSMs for forecasting
  - Combine with TFT: replace LSTM with SSM for sequence encoder
- Add Vintages: use real-time macro data (as available at forecast time) for more realistic evaluation
- Extend dataset: Working Paper (Lustenberger et al., 2026)
  - 10,000+ speeches since 1914, continuously updated

# Conclusion

- in terms of parameters: TFT >> ARIMA > AR
- TFT naturally includes uncertainty bounds
- etc.

# Bibliography

- Lim, B., Arık, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748–1764. <https://doi.org/https://doi.org/10.1016/j.ijforecast.2021.03.012>
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- Shah, A., Paturi, S., & Chava, S. (2023, ). Trillion Dollar Words: A New Financial Dataset, Task & Market Analysis. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=4447632](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4447632)
- Yang, Y., UY, M. C. S., & Huang, A. (2020, ). *FinBERT: A Pretrained Language Model for Financial Communications*. <https://arxiv.org/abs/2006.08097>

# Appendix

# Variable List: Macroeconomic Variables

## 1. Target variables

- CPI, UNRATE, GDP (quarterly: forward-fill missing months)  $\Rightarrow$  AR and ARIMA only take targets

## 2. Covariates

- Monthly variables: payroll (PAYEMS), industrial production (INDPRO), hours worked in manufacturing (AWHMAN), OECD composite leading indicator (USACLI)
- Weekly / daily variables: exchange rates (GBP, YEN), effective federal funds rate (FFR), Fed balance sheet (WALCL)
  - Higher-frequency variables: take mean **and** std per month (volatility)
- Lags at 1, 2, 6, 12 months for all target variables

## 3. Preprocessing: log-difference of CPI, GDP, PAYEMS, INDPRO, AWHMAN

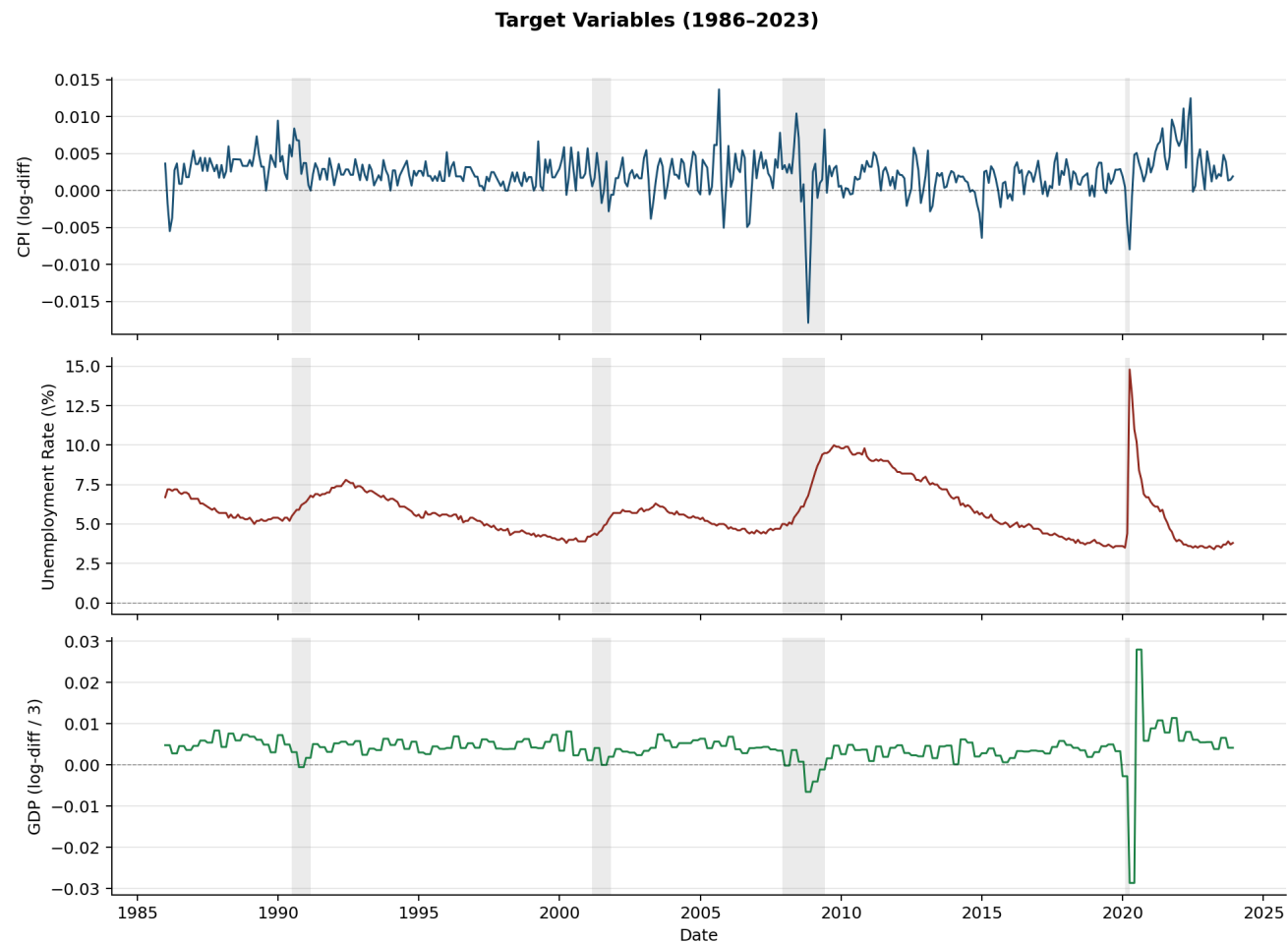
# Data Frame:

| date    | CPI     | PAYEMS  | INDPRO  | UNRATE | AWHMAN  | USACLI  | GDP    | GBP_mean | GBP_std | YEN_mean | YEN_std | FFR_mean |
|---------|---------|---------|---------|--------|---------|---------|--------|----------|---------|----------|---------|----------|
| 1986-01 | 0.0036  | 0.0012  | 0.0055  | 6.7    | -0.0049 | 0.0002  | 0.0048 | 1.4244   | 0.0227  | 199.8905 | 3.6638  | 8.1448   |
| 1986-02 | -0.0018 | 0.0012  | -0.007  | 7.2    | 0.0     | 0.0001  | 0.0048 | 1.4297   | 0.0352  | 184.8516 | 4.586   | 7.86     |
| 1986-03 | -0.0055 | 0.0009  | -0.0071 | 7.2    | 0.0     | -0.0001 | 0.0048 | 1.4674   | 0.0154  | 178.6938 | 1.8252  | 7.479    |
| 1986-04 | -0.0037 | 0.0019  | 0.0014  | 7.1    | -0.0049 | -0.0001 | 0.0028 | 1.4985   | 0.0363  | 175.0918 | 5.1994  | 6.988    |
| 1986-05 | 0.0028  | 0.0013  | 0.0015  | 7.2    | 0.0049  | -0.0002 | 0.0028 | 1.5211   | 0.021   | 167.0314 | 3.4488  | 6.8529   |
| 1986-06 | 0.0037  | -0.0009 | -0.004  | 7.2    | 0.0     | -0.0003 | 0.0028 | 1.5085   | 0.0157  | 167.5419 | 2.5161  | 6.9167   |

| FFR_std | T10Y2Y_mean | T10Y2Y_std | dissent_rate_diss | dissent_net_hawk | dissent_net_hawk_s | n_tighter_sum | n_easier_sum | many_dissent_rec | days_since_fomc | days_to_fomc | fomc_cycle_pos | meeting_this_mon |
|---------|-------------|------------|-------------------|------------------|--------------------|---------------|--------------|------------------|-----------------|--------------|----------------|------------------|
| 1.0103  | 1.0514      | 0.0368     | 0.0922            | 0.25             | 2.0                | 5.0           | 3.0          | 1.0              | 15.0            | 42.0         | 0.2632         | 0.0              |
| 0.1221  | 0.7368      | 0.2171     | 0.0922            | 0.25             | 2.0                | 5.0           | 3.0          | 1.0              | 46.0            | 11.0         | 0.807          | 1.0              |
| 0.2745  | 0.564       | 0.0644     | 0.1131            | 0.0              | 0.0                | 5.0           | 5.0          | 1.0              | 17.0            | 31.0         | 0.3542         | 0.0              |
| 0.2155  | 0.6041      | 0.0734     | 0.1292            | 0.0              | 0.0                | 5.0           | 5.0          | 1.0              | 0.0             | 49.0         | 0.0            | 1.0              |
| 0.085   | 0.6424      | 0.0375     | 0.1131            | 0.0              | 0.0                | 5.0           | 5.0          | 1.0              | 30.0            | 19.0         | 0.6122         | 1.0              |
| 0.175   | 0.6148      | 0.1007     | 0.114             | 0.0              | 0.0                | 5.0           | 5.0          | 1.0              | 12.0            | 38.0         | 0.24           | 0.0              |



# Target Vars

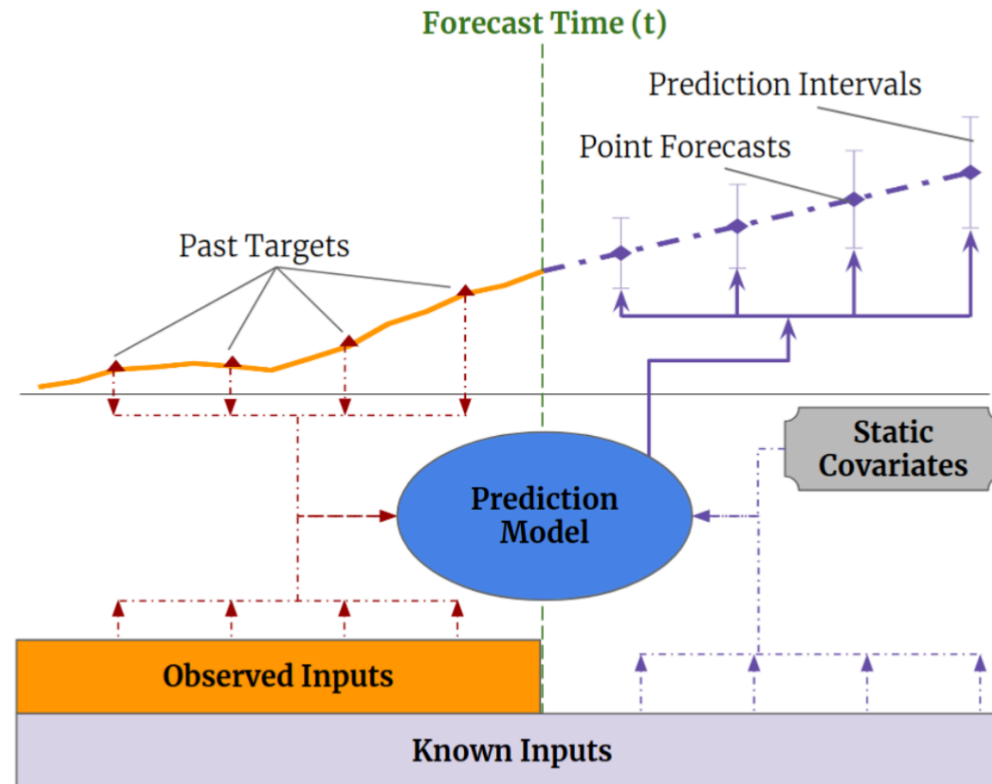


# Variable List

The TFT takes four different types of variables:

1. **Static categoricals:** time-invariant, categorical
  - Identifier and frequency of each time series
2. **Static reals:** time-invariant, continuous
  - Number downloads time series from FRED, how long time series has existed
3. **Time-Varying Known Reals:** time-varying, known
  - Day of week, month, year; if day is holiday
  - FOMC Meeting dates
4. **Time-Varying Unknown Reals:** time-varying, unknown
  - Macroeconomic variables, speech embeddings, FOMC dissents

# TFT: Input Types



# Dimensionality Reduction

FinBERT and FOMC-RoBERTa embeddings are of dimension 768 per speech

⇒ Dimensionality reduction needed for TFT input

We test two methods (number of components  $X$  is hyperparameter, fit only on training data):

1. **PCA**: projects embeddings onto  $X$  principal components
  - 20 components  $\approx$  80% of variance explained
2. **Factor Analysis (FA)**: models embeddings as linear combination of  $X$  latent factors
  - Unlike PCA: explicitly models noise, focuses on **shared** variance
  - Parameters estimated via EM algorithm

Based on your tuning code, here are the two appendix slides filled out: typst

# Hyperparameter Tuning: Stage 1 Search Space

Architecture tuned on macro-only TFT (per target  $\times$  horizon):

| Parameter              | Type        | Range / Values            |
|------------------------|-------------|---------------------------|
| Encoder length         | int         | 12 – 48                   |
| Hidden size            | categorical | {8, 16, 32, 64, 128, 256} |
| Hidden continuous size | categorical | {2, 4, 8, 16}             |
| Dropout                | float       | 0.05 – 0.55               |
| Learning rate          | log-uniform | $10^{-4}$ – 0.15          |
| Normalizer             | categorical | {encoder-none, group}     |

Fixed: `lstm_layers = 2, attention_head_size = 2, max_epochs = 50, patience = 15`

# Hyperparameter Tuning: Stage 2 Search Space

Speech embedding params tuned (architecture fixed from Stage 1):

| Parameter              | Type        | Range / Values           |
|------------------------|-------------|--------------------------|
| Aggregation            | categorical | {mean, decay, attention} |
| Reduction              | categorical | {pca, fa}                |
| PCA components ( $X$ ) | int         | 5 – 30                   |
| Speech window (months) | int         | 3 – 12                   |

Tuned separately for each of 9 combinations: {CPI, GDP, UNRATE}  $\times$  {3, 6, 12} months

# Hyperparameter Tuning: Stage 2 – Optimal Hyperparameters

|        | FOMC-RoBERTa                                                                                       |                                                        |                                                                                                           | FinBERT                                                                                                  |                                                                                                         |                                                                                                       |
|--------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Target | $h=3$                                                                                              | $h=6$                                                  | $h=12$                                                                                                    | $h=3$                                                                                                    | $h=6$                                                                                                   | $h=12$                                                                                                |
| CPI    | Mean<br>PCA, $n = 21$<br>$W = 3$<br>MAE = 0.001847                                                 | Attention<br>FA, $n = 10$<br>$W = 8$<br>MAE = 0.001849 | <b>Attention</b><br><b>PCA, <math>n = 12</math></b><br><b><math>W = 9</math></b><br><b>MAE = 0.001842</b> | <b>Attention</b><br><b>FA, <math>n = 20</math></b><br><b><math>W = 5</math></b><br><b>MAE = 0.001846</b> | <b>Decay</b><br><b>PCA, <math>n = 23</math></b><br><b><math>W = 3</math></b><br><b>MAE = 0.001846</b>   | Mean<br>PCA, $n = 8$<br>$W = 7$<br>MAE = 0.001847                                                     |
| GDP    | <b>Mean</b><br><b>PCA, <math>n = 8</math></b><br><b><math>W = 7</math></b><br><b>MAE = 0.00154</b> | Attention<br>FA, $n = 12$<br>$W = 8$<br>MAE = 0.00171  | Attention<br>FA, $n = 5$<br>$W = 10$<br>MAE = 0.00152                                                     | Attention<br>FA, $n = 6$<br>$W = 12$<br>MAE = 0.00170                                                    | <b>Attention</b><br><b>FA, <math>n = 10</math></b><br><b><math>W = 8</math></b><br><b>MAE = 0.00157</b> | <b>Decay</b><br><b>PCA, <math>n = 20</math></b><br><b><math>W = 12</math></b><br><b>MAE = 0.00151</b> |
| UNRATE | Decay<br>FA, $n = 12$<br>$W = 11$<br>MAE = 1.31293                                                 | Attention<br>FA, $n = 6$<br>$W = 12$<br>MAE = 1.31501  | <b>Mean</b><br><b>PCA, <math>n = 8</math></b><br><b><math>W = 7</math></b><br><b>MAE = 1.11597</b>        | <b>Mean</b><br><b>FA, <math>n = 23</math></b><br><b><math>W = 5</math></b><br><b>MAE = 1.26152</b>       | <b>Mean</b><br><b>PCA, <math>n = 14</math></b><br><b><math>W = 4</math></b><br><b>MAE = 1.30896</b>     | Mean<br>PCA, $n = 25$<br>$W = 4$<br>MAE = 1.20808                                                     |

where  $W$  refers to the speech window size and  $n$  to the optimal number of components for the dimensionality reduction. The best result per target×horizon is shown in bold, and  $h$  stands for forecast horizon.